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GUIDANCE DOCUMENT 5B: SECTOR-SPECIFIC GUIDANCE DOCUMENT ON CHEMICAL SECTOR - HYDROGEN

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1 PREFACE

1.1 Structure and content of the guidance documents

This series of documents has been written to support stakeholders by explaining the requirements of the CBAM Regulation in a non-legislative language. This series of guidance documents should benefit a variety of stakeholders. A specific focus has been put on clarifying the **requirements for operators of installations producing CBAM goods outside of the EU for the definitive period.**

- **Guidance 1** provides an overview of the main changes to the CBAM methodology between the definitive period and the concluded transitional period. It also introduces the CBAM and an overview of the compliance cycle, roles and responsibilities, as well as milestones and deadlines for operators of installations outside the EU during the definitive period and explains the general exemptions from the CBAM.

- **Guidance 2** provides a quick guidance for the intended reader of this document, the operator of an installation producing CBAM goods. It provides a roadmap of the most important concepts of CBAM emissions monitoring and where to find more information in this series of documents.

- **Guidance 3** sets out the monitoring and reporting obligations and recommendations which are potentially applicable to any affected producer of CBAM goods.

- **Guidance 4** provides an overview of how the adjustment for free allocation is to be determined and what information is needed.

- **Guidance documents 5a, 5b, 5c, 5d, 5e and 5f** present an overview per sector of the production processes and value chains for goods that are included in the scope of the CBAM, as well as sector-specific monitoring and reporting considerations for each CBAM good where this is relevant. These sections are supplemented by examples for each sector.

The series of guidance documents is accompanied by an electronic template for information that should be used by installation operators to communicate information to the authorised CBAM declarants and CBAM Registry.

Presentation of numbers in EU documents

To align with EU legal documents, this guidance document uses the following convention when presenting numbers.

The decimal separator used to separate the integral part of a number from its fractional part is a comma, e.g.: 0,890

Thousands, and powers of 10^{3n} thereafter, are separated by a space, e.g.:

- fifteen thousand is written as 15 000
- fifteen million is written as 15 000 000

1.2 Reference regulations and legal acts

Several regulations and legal acts are referred to throughout this series of guidance documents. For ease of reading and to avoid ambiguity, simplified references are used when citing these acts.

- **CBAM Regulation**: Regulation (EU) 2023/956 and Regulation (EU) 2025/2083 amending Regulation (EU) 2023/956. Notably, where article numbers are given in this document without further specification, they always refer to the CBAM Regulation.

  It can be accessed at the following link (consolidated version): **http://data.europa.eu/eli/reg/2023/956/2025-10-20**

- **Methodology Act:** Commission Implementing Regulation (EU) 2025/2547, in legal texts referred to as "Implementing act pursuant to Article 7(7) of the CBAM Regulation".

  It can be accessed at the following link: **http://data.europa.eu/eli/reg_impl/2025/2547/oj**

- **Default Values Act**: Commission Implementing Regulation (EU) 2025/2621, in legal texts referred to as "Implementing act pursuant to Article 7(7) of the CBAM Regulation", also referred to as "default values determined in accordance with Annex IV to the CBAM Regulation".

  It can be accessed at the following link: **http://data.europa.eu/eli/reg_impl/2025/2621/oj**

- **Verification Principles Act**: Commission Implementing Regulation (EU) 2025/2546, in legal texts referred to as "Implementing act pursuant to Article 8(3) of the CBAM Regulation".

  It can be accessed at the following link: **http://data.europa.eu/eli/reg_impl/2025/2546/oj**

- **Accreditation & Verification Requirements Act**: Commission Delegated Regulation (EU) 2025/2551, in legal texts referred to as "Delegated act pursuant to Article 18(3) of the CBAM Regulation".

  It can be accessed at the following link: **http://data.europa.eu/eli/reg_del/2025/2551/oj**

- **Free Allocation Adjustment Act**: Commission Implementing Regulation (EU) 2025/2620, in legal texts referred to as "Implementing act pursuant to Article 31(2) of the CBAM Regulation".

  It can be accessed at the following link: **http://data.europa.eu/eli/reg_impl/2025/2620/oj**

For acronyms and definitions used in this document, please see **Annex B** and **Annex A**.

A series of icons are used throughout to help guide the reader:

Icon	Description of use
	Points to information of particular importance for operators of installations producing CBAM goods.
Simplified!	Highlights simplified approaches of the general requirements of the CBAM.
	Used where recommended improvements are presented.
	Used where other documents, templates or electronic tools are available from other sources.
	Points to examples given for the topics discussed in the surrounding text.

2 CBAM GOODS AND PRODUCTION ROUTES

This section provides guidance on industry-sector-specific rules that apply for the definitive period (from 2026 onwards), for the hydrogen sector. It deals with the specification of products covered by the CBAM and the relevant production routes.

2.1 Foreword to sector-specific sections

The following sections provide an overview of the different production routes for the goods under chemical sectors in CBAM (hydrogen) and provide sector specific guidance. Where goods to which the same functional unit applies are produced in one installation via different production routes, the embedded emissions shall be determined in accordance with the averaging rules applicable in the definitive period.

Additional information on the production processes of the goods can also be found in the BREF[1] documents for best available techniques (BAT).

Diagrams used in the following sections.

For the system boundary graphics presented in the sections below, the **following conventions** are applied:

- **Production processes** (for which monitoring of the direct emissions would take place) are shown as **rectangles with white background**.
- **Optional** processes (e.g. CCS/CCU) are shown in **blue** boxes. In particular, CCS/CCU would not be taken into account for developing default values, but where you (as an operator) use them, the related emissions or emission savings should be taken into account for determining actual embedded emissions in line with the CBAM Regulation and the Methodology Act.
- **Materials** which are considered to have no embedded emissions are shown in **pink** boxes, materials with embedded emissions (precursor materials and final products, i.e. goods under the CBAM) in **green** boxes. Simple goods are shown in **normal** font, complex goods in **bold** font. Materials which are **optional** are shown in boxes with a **blue dashed** outline and must be included where actually used at the installation.
- Input materials are presented without trying to be complete. This means that the focus is on materials which are relevant for demonstrating the differences between different production routes. As a consequence, less important input materials and in particular fuels are usually omitted in order to keep graphs simple.
- **Electricity** as input is shown only in cases where it is the main "precursor" of the process (i.e. in particular for electric arc furnaces and electrolysis processes). Electricity is shown in **light grey** boxes.

[1] BAT Reference document (BREF), BAT being "Best Available Techniques" as defined by the IED (Industrial Emissions Directive). Relevant BREF documents are those for: the production of cement; for iron and steel production; large volume inorganic chemicals (which includes fertilisers); for Chlor-alkali; and for non-ferrous metals (which includes both aluminium and ferro-alloys). All BREFs can be found at the European IPPC Bureau, under https://eippcb.jrc.ec.europa.eu/reference.

2.2 Identifying CBAM goods

This section explains how goods covered by the CBAM are defined and identified in the Regulation. The textbox below signposts the key sections for the definition and reporting of CBAM goods, relevant for the CBAM transitional period.

Methodology Act references:

- **Annex I,** point 2 Table 1 "Mapping of CN codes to aggregated goods categories"
- **Annex III** "Rules for attributing emissions to goods".

2.3 Chemicals sector – Hydrogen

The textbox below signposts sector-specific sections in the Methodology Act, relevant for the CBAM definitive period.



Methodology Act references:

- **Annex I,** Table 1 "Mapping of CN codes to aggregated goods categories" (aggregated goods category "Hydrogen") of the Methodology Act adopted pursuant to Article 7(7) of Regulation (EU) 2023/956.
- **Annex I,** point 3.6 "Hydrogen", which defines the functional unit, system boundaries and precursors for hydrogen production routes, and
- **Annex III** to the Methodology Act, which sets out the rules for attributing emissions of an installation to goods.

2.3.1 Unit of production and embedded emissions

The quantity of hydrogen imported into the EU must be expressed in metric tonnes (as pure hydrogen). As an operator, you must record the quantity of hydrogen produced by installation or production process, for the purposes of reporting and for use in the calculation of specific embedded emissions in accordance with the CBAM Regulation and the Methodology Act.

Industrial sector	**Chemicals – Hydrogen**
Production unit of goods	Tonnes (metric) pure hydrogen, reported separately by installation or production process in the country of origin
Associated activities	Producing hydrogen by steam reforming or partial oxidation of hydrocarbons, water electrolysis, Chlor-Alkali electrolysis or production of sodium chlorate, steam cracking of hydrocarbons.
Relevant greenhouse gases	Carbon dioxide (CO_2)
Direct Emissions	Tonnes (metric) of CO_2e

Industrial sector	Chemicals – Hydrogen
Indirect Emissions	Quantity of electricity consumed (MWh), source and emissions factor used to calculate the indirect emissions in Tonnes (metric) of CO_2 or CO_2e. Indirect emissions were monitored and reported during the transitional period but, for the definitive period, hydrogen is included in Annex II to the CBAM Regulation ("Goods for which only direct emissions are taken into account"), so indirect emissions are not taken into account in the embedded emissions of hydrogen for CBAM purposes.
Unit for embedded emissions	Tonnes CO_2e emissions per tonne of good, reported separately for each type of good, by installation in the country of origin

In the definitive period, for CBAM purposes the hydrogen sector has to account for direct emissions only, since hydrogen is listed in Annex II to the CBAM Regulation as a good for which only direct emissions are taken into account. Indirect emissions were reported separately during the transitional period but are not included in the embedded emissions of hydrogen in the definitive period[2]. Emissions must be reported in metric tonnes CO_2 equivalent (t CO_2e) emissions per tonne of output. This figure must be calculated for the specific installation or production process in your country of origin.

The following sections set out how the system boundaries of different hydrogen production routes should be defined and identify elements of the production process that must be included for the purposes of monitoring and reporting.

2.3.2 Definition and explanation of sector CBAM goods covered

The following Table 2-1 lists the relevant goods in scope for the CBAM definitive period in the hydrogen industry sector. The CN codes in the central column are the starting point for which 'production processes' are to be defined for the purpose of monitoring at installation level in a manner consistent with the system boundaries and production routes set out in Annex I to the Methodology Act.

Table 2-1: CBAM goods in the chemicals sector – hydrogen[3]

Aggregated goods category	Product CN Code	Description
Hydrogen	2804 10 00	Hydrogen

[2] For this sector indirect emissions were reported during the transitional period, but hydrogen is listed in Annex II to Regulation (EU) 2023/956 as a good for which only direct emissions are taken into account, so indirect emissions are not taken into account in embedded emissions in the definitive period.

[3] Source: Annex I to the CBAM Regulation; Annex I ("Mapping of CN codes to aggregated goods categories") to the Methodology Act.

Hydrogen is defined as a simple good in accordance with Article 1 and Annex I of the Methodology Act, as there are no possible precursors listed for the production of hydrogen and the embedded emissions are determined solely from the direct emissions of the hydrogen production process.

There are **no precursors** relevant for hydrogen. However, hydrogen may itself be a precursor for other processes, where it is separately produced for use as a chemical feedstock to produce ammonia, or to produce pig iron or direct reduced iron (DRI).

Where hydrogen is used as a precursor in the production of ammonia and the conditions laid down in the Methodology Act for disregarding indirect emissions of precursors are met (e.g. where the precursor itself is listed in Annex II), the indirect emissions of electricity associated with the hydrogen precursor may be disregarded when determining the embedded emissions of the ammonia CBAM goods.

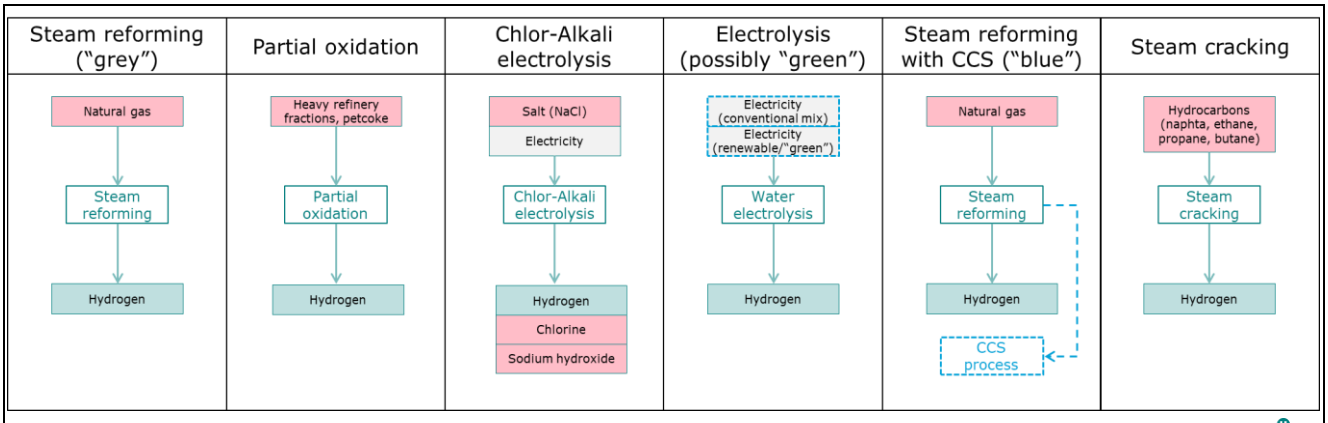
The production of hydrogen is by a number of different production routes, outlined below.

2.3.3 Definition and explanation of relevant production processes and routes

Hydrogen can be produced from various feedstocks including plastic wastes, but currently it is derived mostly from fossil fuels. Hydrogen production units are typically integrated into larger industrial processes e.g. as for an installation producing ammonia. For CBAM purposes, the relevant production routes and their system boundaries are defined for the hydrogen aggregated goods category in Annex I, point 3.6 to the Methodology Act, and emissions are attributed to hydrogen in accordance with Annex III to the Methodology Act.

The following diagram illustrates the variety of different routes by which hydrogen may be produced. Note that chlor-alkali electrolysis and water electrolysis are not mentioned in the Methodology Act, as these production routes include minimal direct emissions.

Figure 2-1: Overview of different production routes for hydrogen relevant for CBAM



Notes to figure: For hydrogen (Annex II goods), only direct emissions are taken into account for embedded emissions; indirect emissions from electricity are not included.

The system boundaries for direct emissions monitoring for hydrogen include all processes directly or indirectly linked to hydrogen production, and all fuels used in the production of hydrogen, in line with Annex I, point 3.6 and Annex III to the Methodology Act.

Note that although other production routes for hydrogen are possible, only the production of pure hydrogen or mixtures of hydrogen with nitrogen usable in ammonia production shall be considered for the purposes of the hydrogen aggregated goods category under Annex I to the CBAM Regulation. Not covered are the production of synthesis gas or of hydrogen within refineries or organic chemical installations, where the hydrogen is exclusively used within those plants, and is not used for the production of goods under the CBAM Regulation and is therefore outside the scope of the hydrogen CBAM good.

2.3.3.1 Hydrogen – Steam reforming and partial oxidation production routes

The fossil gas (methane) feedstock for this process is converted to carbon dioxide and hydrogen through primary and secondary steam reformation. The overall reaction is highly endothermic and process heat is supplied by the combustion of natural gas or other gaseous fuel. Carbon monoxide produced is almost all converted to carbon dioxide by the process.

The Methodology Act, in particular Annex I, point 3.6.2.1 ("Steam reforming and partial oxidation"), defines the system boundaries for direct emissions monitoring for the steam reforming (and partial oxidation[4]) production routes. For the purposes of this guidance, those system boundaries can be described as encompassing:

> *" – All processes directly or indirectly linked to hydrogen production and the separation of hydrogen and carbon monoxide, and flue gas cleaning;*
>
> *– All fuels used in the hydrogen production process irrespective of their energetic or non-energetic use, and fuels used for other combustion processes including for the purpose of producing hot water or steam."*

There are no precursors for this production process. In the definitive period, hydrogen is listed in Annex II to the CBAM Regulation as a good for which only direct emissions are taken into account. Indirect emissions resulting from electricity consumed by the production process are therefore not taken into account in the embedded emissions of hydrogen for CBAM purposes, although operators may still monitor such data for other regulatory or internal purposes.

In line with the above definition of systems boundaries, the following production steps may be regarded as being within the system boundaries of a hydrogen (steam reformation) installation:

- Raw material pre-treatment – natural gas desulphurisation
- Steam reformation – primary and secondary, H_2/CO generation

[4] The partial oxidation production route differs from steam reforming in that heavier hydrocarbon fractions such as residual fuel oil or coal are used as feedstocks to produce hydrogen. The system boundaries for the partial oxidation production route are very similar to those for the steam methane reformation production route, and in addition include the generation of oxygen as a process gas, and gasification to produce a synthesis gas, which is then cleaned to remove soot and sulphur impurities, from which CO / CO_2 are removed to obtain H_2.

- Shift conversion – carbon monoxide to carbon dioxide and hydrogen

- Separation & purification – CO_2 removal, separation processes as present including cryogenic, adsorption, absorption, membrane, hydrogenation (methanation)

- Emissions control – for treating releases to air, water or ground

The stream of carbon dioxide produced by the steam reforming process is very pure and is separated and captured for further use, e.g. for urea production. A variation on this process may be with permanent geological storage i.e. carbon capture and sequestration (CCS). Where CO_2 is captured and permanently stored or chemically bound in accordance with the CBAM rules, this is reflected in the determination of embedded emissions in line with the Methodology Act.

2.3.3.2 Hydrogen – Steam cracking production route

Hydrogen is produced as a co-product of the thermal cracking of hydrocarbons using feedstocks such as naphtha and natural gas condensates (i.e. ethane, propane and butane) alongside the simultaneous production of light olefins such as of ethylene and propylene.

The Methodology Act, in particular Annex I, point 3.6.2.2 ("Steam cracking"), defines the system boundaries for direct emissions monitoring of the steam cracking production route. For the purposes of this guidance, the system boundaries for direct emissions monitoring can be described as encompassing:

> *" – All processes directly linked to hydrogen production;*
>
> *– All processes directly or indirectly linked to the production processes, and from flue gas cleaning."*

There are no precursors for this production process. As for steam methane reforming, in the definitive period only direct emissions are taken into account in the embedded emissions of hydrogen under the CBAM. Indirect emissions from electricity consumption are not taken into account in the embedded emissions of hydrogen, although operators may still monitor such data for other regulatory or internal purposes.

In line with the above definition of systems boundaries, the following production steps may be regarded as being within the system boundaries of a hydrogen (steam cracking) installation:

- Production of hydrogen as a co-product of the steam cracking of hydrocarbons – mixing of hydrocarbon feedstocks with steam, cracking of hydrocarbons in furnaces, generation of hydrogen as a by-product, and collection.

- Gas cooling, drying and purification – removal of water vapour and other impurities from hydrogen gas.

- Emissions control – for treating releases to air, water or ground.

2.3.3.3 Hydrogen – Electrolysis of water production route

Water electrolysis is a standalone, non-integrated production process that produces a very pure stream of hydrogen gas. Direct emissions from this process are minimal. Electricity consumption is the main driver of greenhouse gas emissions for this route, and the associated emissions are treated as indirect emissions from electricity generation. However, in the definitive period, those indirect emissions are not taken into account in the embedded emissions of hydrogen for CBAM purposes, as hydrogen is listed in Annex II to the CBAM Regulation as a good for which only direct emissions are taken into account pursuant to Article 3(2) to the Methodology Act.

The Methodology Act removes the route-specific provisions for the electrolysis of water previously listed in section 3.6.2 of the transitional methodology. Consequently, this route is now governed by the generic cross-sectoral rules in Annex I, Section 3.1 and the special provisions for hydrogen in Annex I, Section 3.6.1. For the purposes of this guidance, where relevant, the system boundaries for direct emissions monitoring are defined by the general rule in Section 3.1 as encompassing:

" – all processes directly or indirectly linked to the production processes."

There are no precursors for this production process under Article 1(7), as input materials like water are not CBAM goods listed in Annex I to the CBAM Regulation. Furthermore, because this is a multifunctional chemical process yielding hydrogen and co-product oxygen, operators must attribute emissions based on the molar ratio rule defined in Annex III, Section A.2.1 using equation 43.

During the transitional period pursuant to Implementing Regulation (EU) 2023/1773, indirect emissions resulting from electricity consumed by the production process were monitored and, where the produced hydrogen was certified in accordance with Commission Delegated Regulation (EU) 2023/1184, an emission factor of zero for the electricity could be used, with the applicable rules on indirect embedded emissions in that Regulation applying in other cases. In the definitive period, however, hydrogen is listed in Annex II to the CBAM Regulation as a good for which only direct emissions are taken into account pursuant to Article 3(2) of this Regulation, so indirect emissions from electricity consumption are not taken into account in the embedded emissions of hydrogen for CBAM purposes.

2.3.3.4 Hydrogen – Chlor-alkali electrolysis (and production of chlorates) production routes

Hydrogen is produced as a co-product of the electrolysis of brine, alongside the simultaneous production of chlorine and sodium hydroxide. There are three basic chlor-alkali process techniques: mercury cell, diaphragm cell and the membrane cell. All three cell techniques produce hydrogen, which is formed at the cell cathode and which leaves the cell at very high purity. The hydrogen gas produced is cooled, dried and purified to remove water vapour, and other impurities, which may in some cases include oxygen, and is then compressed and stored or exported off site.

The Methodology Act removes the route-specific provisions for the Chlor-alkali electrolysis (and production of chlorates) previously listed in section 3.6.2 of the transitional methodology. Consequently, this route is now governed by the generic cross-

sectoral rules in Annex I, Section 3.1 and the special provisions for hydrogen in Annex I, Section 3.6.1. For the purposes of this guidance, where relevant, the system boundaries for direct emissions monitoring are defined by the general rule in Section 3.1 as encompassing:

" – all processes directly or indirectly linked to the production processes."

There are no precursors for this production process under Article 1(7), as input materials are not CBAM goods listed in Annex I to the CBAM Regulation. Furthermore, because this is a multifunctional chemical process yielding chlorine, sodium hydroxide and co-product hydrogen, operators must attribute emissions based on the molar ratio rule defined in Annex III, Section A.2.1.

During the transitional period, indirect emissions that resulted from electricity consumed by the production process also needed to be monitored, and where the produced hydrogen had been certified to comply with Commission Delegated Regulations (EU) 2023/1184 (1), an emission factor of zero for the electricity could be used, with the applicable rules on indirect embedded emissions in that Regulation applying in other cases. In the definitive period, however, hydrogen is listed in Annex II to the CBAM Regulation as a good for which only direct emissions are taken into account pursuant to Article 3(2) of this Regulation, so indirect emissions from electricity consumption are not taken into account in the Sector specific monitoring and reporting

3 CHEMICALS – HYDROGEN SECTOR-SPECIFIC REQUIREMENTS FOR MONITORING AND REPORTING

The textbox below signposts sector-specific sections in the Methodology Act, relevant for the CBAM definitive period.

Methodology Act references:

- **Annex I, point 3** – Functional unit and system boundaries. Sub-section **3.6** (Hydrogen).
- **Annex III, Section A.2.1** – Chemicals and fertilisers
- **Annex IV, point 2** – Sector-specific parameters to be included in the operator's emissions report.

3.1 Sector-specific requirements for monitoring and reporting

Direct embedded emissions should be monitored in line with the methodology set out in the Methodology Act.

Exclusions

As an operator, you should note that only the production of pure hydrogen or mixtures of hydrogen with nitrogen usable in ammonia production shall be considered. Not covered are the production of synthesis gas or of hydrogen as a precursor within refineries or organic chemical installations, where the hydrogen is exclusively used within those plants and not used for the production of goods under the CBAM Regulation.

3.1.1 Emissions monitoring

Relevant emissions that must be monitored and reported for the hydrogen sector are:

- Carbon dioxide emissions (direct) from the fuel combustion process in the hydrogen or synthesis gas production process, primary and secondary steam reforming of natural gas, or partial oxidation other hydrocarbons; from stationary plant only (excludes emissions from any mobile units such as vehicles).

- Carbon dioxide emissions (direct) resulting from the production of measurable heating (for the purpose of producing hot water or steam) and cooling that is consumed within the system boundaries of the production process, regardless of the location of the production of the heat (i.e. from on-site generation or from imports from off-site).

- Carbon dioxide emissions (direct) from electrolysis are minimal and so where these emissions are significant they are likely to result from ancillary plant.

- Carbon dioxide emissions (direct) resulting from emissions control (e.g. from carbonate raw materials such as soda ash used for acidic flue gas cleaning).

Direct emissions from the different source streams above are not reported separately but are added together to result in the total direct emissions for the installation or production process.

In the definitive period, indirect emissions from electricity consumed do not have to be reported.

3.1.2 Additional rules

Attribution of emissions where different products are simultaneously produced

Additional rules apply for the attribution of direct (and where applicable indirect) emissions to the different products of the following production processes that fall under the chemicals aggregated goods category. Where these products have been produced simultaneously as co-products of a single multifunctional production process, the attribution among the different co-products (whether or not they are CBAM goods) shall be based on molar ratio, with the generic formula:

$$Em_i = Em_{total}\left(\frac{\frac{m_{i,prod}}{M_i}}{\sum_{j=1}^{n}\frac{m_{j,prod}}{M_j}}\right) \text{ (Equation 43)}$$

Where:

Em_i … Either direct or indirect emissions attributed to each co-product *i* produced over the reporting period, expressed in tonnes of CO_2

Em_{total} … Either direct or indirect emissions of the whole (multifunctional) production process over the reporting period, expressed in tonnes of CO_2

$m_{i,prod}$ …Mass of the co-product *i* under consideration, produced in the installation over the reporting period, expressed in tonnes

$m_{j,prod}$ …Mass of each co-product *j* produced in the installation over the reporting period, expressed in tonnes

M_i … Molar mass of the co-product *i* under consideration

M_j … Molar mass of each co-product *j*

If the molar mass of one of the co-products is not known, the emissions will be attributed based on mass of the co-products. In that case, the formula would be as follows:

$$Em_i = Em_{total}\left(\frac{m_{i,prod}}{\sum_{j=1}^{n} m_{j,prod}}\right)$$

Worldwide, the most important route for hydrogen production is steam reforming and gasification of coal, where hydrogen is the main product. However, equation 43 is relevant in particular for production routes of hydrogen where hydrogen is not the main product. Taking into account the production routes listed in Annex I to the Methodology Act, this is in particular the case for steam cracking. Depending on the raw material used in the cracker (e.g. naphtha, ethane, etc.) different product mixes are found. The operator should then define which of the co-products are 'high value chemicals' (HVCs), i.e. products that will be used for economic purposes, while other co-products are usually not purified for chemical use, but used as fuel in various parts of the installation. Hence, the molar mass and mass produced of each of the HVCs will have to be used in the equation above. Regarding the steam cracker, a useful approach is to use the scope of HVC aligned with the EU ETS Free Allocation Rules which define the following HVCs as relevant for the steam cracker product benchmark: Hydrogen (H_2), acetylene (C_2H_2), ethylene (C_2H_4), propylene (C_3H_6), butadiene (C_4H_6) and benzene (C_6H_6). Furthermore, where those co-products are separated and used, also the following may be relevant: Toluene, Xylenes, Butenes.

3.1.3 Processes that have only indirect emissions

For hydrogen production, production routes exist that have only indirect emissions. However, those are currently not relevant for hydrogen as CBAM good, as hydrogen is listed in Annex II to the CBAM Regulation. Nevertheless, where an operator uses hydrogen from such sources as precursor, at least the respective production route must be named, and for use of actual values (i.e. zero), verification in line with CBAM rules of the installation producing the hydrogen will be required. However, using a risk-based approach, verifiers may consider a simplified approach for hydrogen free of direct emissions. For example, where hydrogen is certified as an RFNBO (Renewable Fuel of Non-Biological Origin) under the Renewable Energy Directive (RED II)[5], verification can be strongly simplified. Such certification is required for the purpose of importing "green hydrogen" under the EU's renewable energy framework.

Furthermore, specific cases for co-product attribution exist, although they would only be relevant if indirect emissions were relevant for hydrogen. They are mentioned here for reasons of completeness. Such calculation was necessary in the transitional phase.

[5] In accordance with Commission Delegated Regulation (EU) 2023/1184 supplementing Directive (EU) 2018/2001 […] by establishing a Union methodology setting out detailed rules for the production of renewable liquid and gaseous transport fuels of non-biological origin. See http://data.europa.eu/eli/reg_del/2023/1184/oj

- Electrolysis of water – where oxygen is released to the atmosphere, all emissions from the production process are attributed to the hydrogen product. However, if oxygen is collected and used in other production processes or is sold then molar proportions are used to attribute emissions, using equation 43.

- Chlor-Alkali electrolysis and the production of chlorates – molar proportions are used to attribute emissions to the hydrogen produced, using equation 43.

3.1.4 Additional reporting requirements

The following Table 3-1 lists out the additional information that needs to be provided by you as an operator to importers, in your emissions data communication to them.

Table 3-1: Additional chemical sector parameters requested in the CBAM declaration

Aggregated good category	Reporting requirement in the annual report
Hydrogen	– None

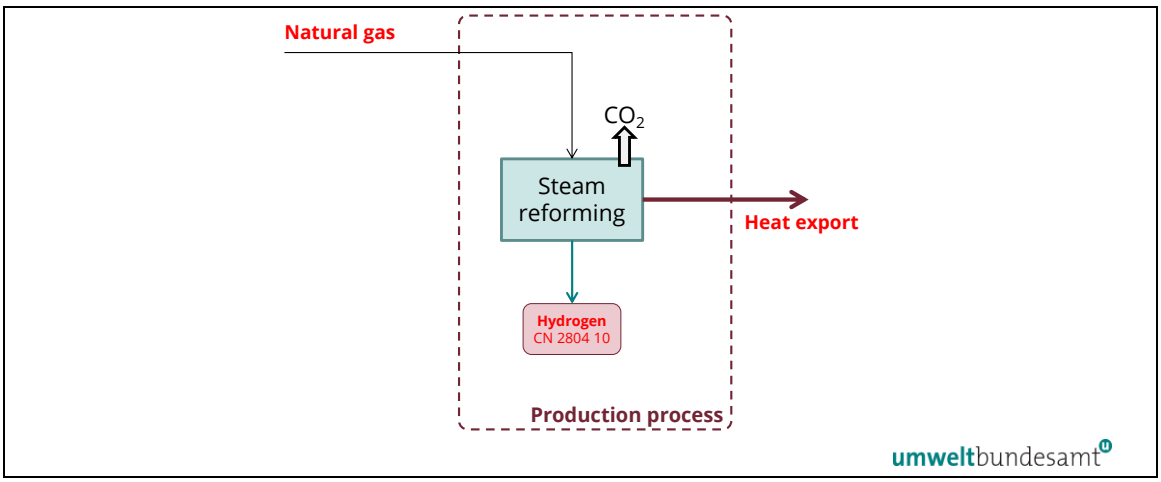
4 WORKED EXAMPLE FOR THE HYDROGEN SECTOR

The following worked example shows how specific embedded emissions are derived for hydrogen produced by the steam reforming production route.

The resulting embedded emissions of imports into the EU are then calculated at the end of the example for reporting in the definitive period.

The diagram below gives an outline view of the installation and shows the system boundaries as a hatched line for the single production process. The physical units carrying out the production process have been grouped under 'steam reforming' and the inputs and outputs and sources of emissions have been identified.

Figure 4-1: Hydrogen example – Overview and complete monitoring approach for hydrogen



A single production process is defined for steam reforming. Inputs are natural gas (both as a raw material / feedstock for the process, and as a fuel). Outputs are hydrogen product and heat exported to other parts of the installation or to a district heating network.

The inputs and outputs highlighted in red text in Table 4-1 are the parameters that would need to be monitored by the operator in order to attribute emissions and determine direct specific embedded emissions for the production process.

The direct emissions that are monitored in this example result from:

- Direct emissions from fuel combustion and from the steam reforming process[6].
- For the purpose of calculating attributed emissions of the process, an equivalent of emissions associated with the heat export needs to be determined, and subtracted from attributed emissions. The latter section explains that for cases of heat of unknown origin, as a default, natural gas with a boiler efficiency of 90% is to be used, if no other data sources can be used. This gives a value of 62,3 t CO_2/TJ.

The activity level of the hydrogen good produced also needs to be monitored.

Table 4-1 summarises the inputs and outputs from the process that are monitored for determining total direct specific embedded emissions.

Table 4-1: Example calculation of total direct emissions attributed to hydrogen net of emissions for a heat export.

Direct emissions	**AD (t)**	**NCV (GJ/t)**	**Energy (TJ)**	**EF (t CO_2/TJ)**	**Emissions (t CO_2)**
Input natural gas	190 000	48	9 120	56,1	511 632
Heat export			-800	62,3	-49 867
Total direct emissions of the installation					**461 765**

Total direct emissions of the installation result from one single source stream (natural gas). There is no need to distinguish combustion and process emissions for this purpose. In this example, these are wholly attributed to the hydrogen product, net of emissions attributed to a heat export. If the almost pure CO_2 produced by this process were to be captured and transferred to a geological CO_2 storage site, the related emissions could be deducted, provided the receiving installation carries out monitoring under the CBAM.

Table 4-2: Calculation of embedded emissions of hydrogen product (example)

Product	**Activity level (t)**	**Process total emissions (t CO_2)**	**SEE (t CO_2 / t H_2)**
Hydrogen	55 000	461 765	**8,396**

[6] Emissions of carbon monoxide (CO) to the atmosphere from the process are not counted as outgoing source stream in the mass balance but are considered as the molar equivalent amount of CO_2 emissions.

Free allocation adjustment

Hydrogen is a simple good with no differentiation of the EU ETS benchmark for any factors such as production routes. Therefore, the specific free allocation results directly from a look-up of the respective CBAM benchmark from the Annex of the Free Allocation Adjustment Act using the CN code for hydrogen: 5,089 t CO_2 / t H_2.

- Assumption: Year = 2028.:
  - CSCF = 1,00
  - CBAM = 90%

SEFA = 5,089 × 0,90 × 1,00 = **4,580 t CO_2 / t H_2**

CBAM obligation upon import:

Using the above approach, the CBAM reporting obligation for the import of hydrogen product into the EU during the definitive period may then be determined; for example, for the import of 100 tonnes of hydrogen product in year 2028 produced by methane steam reforming:

Definitive period:

- Total direct embedded emissions = 100 t × 8,396 t / t CO_2 = **839,6 t CO_2**

  Minus Free Allocation adjustment: **458,0 t CO_2**

  (Further deduction could be made if a carbon price was paid in the country of production – outside the scope of this guidance).

- **Total CBAM obligation** to be covered by the authorised declarant with CBAM certificates: 839,6 – 458,0 = **381,6 certificates**

Annex A List of definitions

Term	Definition
'Accuracy'	means the closeness of the agreement between the result of a measurement and the true value of the particular quantity or a reference value determined empirically using internationally accepted and traceable calibration materials and standard methods, taking into account both random and systematic factors
'Activity data'	means the amount of fuels or materials consumed or produced by a process relevant for the calculation-based methodology, expressed in terajoules (TJ), mass in tonnes or (for gases) volume in normal cubic metres, as appropriate
'Actual emissions'	means the emissions calculated based on primary data from the production processes of goods and from the production of electricity consumed during those processes as determined in accordance with the methods set out in Annex IV [to the Methodology Act]
'Activity level'	means the quantity of goods to which the same functional unit applies that are produced within the system boundaries of a production process during a reporting period
'Aggregated goods categories'	means aggregated goods categories pursuant to Table 1 of point 2 of Annex I. A group of CBAM goods with their CN codes, used for the purpose of defining system boundaries of production processes.
'Agricultural, aquaculture, fisheries and forestry residues'	means residues that are directly generated by agriculture, aquaculture, fisheries and forestry and that do not include residues from related industries or processing
'Authorised CBAM declarant'	means a person authorised by the competent authority in accordance with Article 17 [of the CBAM Regulation]. The 'authorised CBAM declarant' [which replaces the term 'reporting declarant' from the transitional period] may be importers of CBAM goods, or their indirect customs representatives, who have applied for and been granted the status of authorised CBAM declarant.
'Batch'	means an amount of fuel or material representatively sampled and characterised, and transferred as one shipment or continuously over a specific period of time
'Biomass'	means biomass as defined in Article 2, point (24) of the Renewable Energy Directive (RED II)[7]; it includes bioliquids and biofuels as defined in Article 2, point (32) and (33), biomass fuels as defined in Article 2, point (27) and biogas as defined in Article 2 point (28) of the RED II;

[7] Directive (EU) 2018/2001

Term	Definition
'Biomass fraction'	means the ratio of carbon stemming from biomass to the total carbon content of a fuel or material, expressed as a fraction
'Calculation factors'	means net calorific value, emission factor, preliminary emission factor, oxidation factor, conversion factor, carbon content or biomass fraction
'Calibration'	means the set of operations, which establishes, under specified conditions, the relations between values indicated by a measuring instrument or measuring system, or values represented by a material measure or a reference material and the corresponding values of a quantity realised by a reference standard
'Carbon price'	means the monetary amount paid in a third country, under a carbon emissions reduction scheme, in the form of a tax, levy or fee or in the form of emission allowances under a greenhouse gas emissions trading system, calculated on greenhouse gases covered by such a measure, and released during the production of goods
'CBAM certificate'	means a certificate in electronic format corresponding to one tonne of CO_2e of embedded emissions in goods
'CO_2 emission factor'	means the weighted average of the CO_2 intensity of electricity produced from fossil fuels within a geographic area. The CO_2 emission factor is the result of the division of the CO_2 emission data of the electricity sector by the gross electricity generation based on fossil fuels in the relevant geographic area. It is expressed in tonnes of CO_2 per megawatt-hour
'Combined nomenclature' (CN)	means the classification of goods, designed to meet the needs of: i) the Common customs tariff, setting import duties for products imported into the European Union (EU), as well as the Integrated tariff of the European Communities (Taric), incorporating all EU and trade measures applied to goods imported into and exported out of the EU; ii) the international trade statistics of the EU. The CN provides the means of collecting, exchanging and publishing data on EU international trade statistics. It is also used for the collection and publication of international trade statistics in intra-EU trade.[8]
'Combustion emissions'	means greenhouse gas emissions occurring during the exothermic reaction of a fuel with oxygen
'Competent authority'	means the authority designated by each Member State in accordance with Article 11 [of the CBAM Regulation]

[8] For definition see: https://ec.europa.eu/eurostat/statistics-explained/index.php?title=Glossary:Combined_nomenclature_(CN)

Term	Definition
'Continuous emission measurement' (CEM)	means a set of operations having the objective of determining the value of a quantity by means of periodic measurements, applying either measurements in the stack or extractive procedures with a measuring instrument located close to the stack, whilst excluding measurement methodologies based on the collection of individual samples from the stack
'Control system'	means the operator's risk assessment and entire set of control activities, including the continuous management thereof, that an operator has established, documented, implemented and maintained pursuant to point A.2. of Annex II
'Complex goods'	means goods other than simple goods
'Conservative'	means that a set of assumptions is defined in order to ensure that no under-estimation of reported emissions or over-estimation of production of heat, electricity or goods occurs
'Conversion factor'	means the ratio of carbon emitted as CO_2 to the total carbon contained in the source stream before the emitting process takes place, expressed as a fraction, considering CO emitted to the atmosphere as the molar equivalent amount of CO_2
'Co-product'	means any of two or more products resulting from the same production process
'Customs declarant'	means a declarant as defined in Article 5, point (15) of Regulation (EU) No 952/2013 lodging a customs declaration for release for free circulation of goods in its own name or the person in whose name such a declaration is lodged
'CCUS system'	means a group of economic operators with technically connected installations and transport equipment for CO_2 capture, transport, use in production of goods, or geological storage
'Data flow activities'	mean activities related to the acquisition, processing and handling of data that are needed to draft an emissions report from primary source data
'Data set'	means one type of data, either at installation level or production process level as relevant in the circumstances, as any of the following: (a) the amount of fuels or materials consumed or produced by a production process as relevant for the calculation-based methodology, expressed in terajoules, mass in tonnes, or for gases as volume in normal cubic metres, as appropriate, including for waste gases; (b) a calculation factor;

Term	Definition
	(c) net quantity of measurable heat, and the relevant parameters required for determining this quantity, in particular: i) mass flow of heat transfer medium; and ii) enthalpy of transmitted and returned heat transfer medium, as specified by composition, temperature, pressure and saturation; (d) quantities of non-measurable heat, specified by the relevant quantities of fuels used for producing the heat, and the net calorific value (NCV) of the fuel mix; (e) quantities of electricity; (f) quantities of CO_2 transferred between installations; (g) quantities of precursors received from outside the production process, and their relevant parameters, such as country of origin, used production route, specific direct and indirect emissions;
'Default value'	means a value, which is calculated or drawn from secondary data, which represents the embedded emissions in goods
'Direct emissions'	means emissions from the production processes of goods including emissions from the production of heating and cooling that is consumed during the production processes, irrespective of the location of the production of the heating or cooling
'Embedded emissions'	means direct emissions released during the production of goods and indirect emissions from the production of electricity that is consumed during the production processes, calculated in accordance with the methods set out in Annex IV and further specified in the implementing acts adopted pursuant to Article 7(7)
'Emissions'	mean the release of greenhouse gases into the atmosphere from the production of goods
'Emission factor'	means the average emission rate of a greenhouse gas relative to the activity data of a source stream assuming complete oxidation for combustion and complete conversion for all other chemical reactions
'Emission factor' for electricity	means the default value, expressed in CO_2e, representing the emission intensity of electricity consumed in production of goods.
'Emission source'	means a separately identifiable part of an installation or a process within an installation, from which relevant greenhouse gases are emitted
EU ETS	means the system for greenhouse gas emissions allowance trading within the Union in respect of activities listed in Annex I to Directive 2003/87/EC other than aviation activities

Term	Definition
'EU ETS allowance'	means an allowance as defined in Article 3, point (a), of Directive 2003/87/EC in respect of activities listed in Annex I to that Directive other than aviation activities
'Fossil carbon'	means inorganic and organic carbon that is not biomass
'Fossil fraction'	means the ratio of fossil carbon to the total carbon content of a fuel or material, expressed as a fraction
'Fugitive emissions'	means irregular or unintended emissions from sources that are not localised, or too diverse or too small to be monitored individually
'Functional unit'	means the reference unit used for the calculation of embedded emissions in goods
'Goods'	means goods listed in Annex I [to the CBAM the CBAM Regulation]
'Greenhouse gases'	mean greenhouse gases as specified in Annex I [to the CBAM the CBAM Regulation] in relation to each of the goods listed in that Annex
'Heat meter'	means a thermal energy meter or any other device to measure and record the amount of thermal energy produced based upon flow volumes and temperatures
'Importer'	means either the person lodging a customs declaration for release for free circulation of goods or a bill of discharge in accordance with Article 175(5) of Delegated Regulation (EU) 2015/2446 in its own name and on its own behalf or, where the customs declaration is lodged by an indirect customs representative in accordance with Article 18 of Regulation (EU) No 952/2013, the person on whose behalf such a declaration is lodged
'Importation'	means release for free circulation as provided for in Article 201 of Regulation (EU) No 952/2013
'Indirect customs Representative'	A customs representative appointed by an importer established in a Member State. If the importer appoints an indirect customs representative and they agree, they can act as an authorised CBAM declarant and submit the application for an authorisation on behalf of the importer. The indirect customs representative is liable for all operations made for the importers it represents
'Indirect emissions'	means emissions from the production of electricity which is consumed during the production processes of goods, irrespective of the location of the production of the consumed electricity
'Inherent CO_2'	means CO_2 which is part of a source stream.
'Installation'	means a stationary technical unit where a production process is carried out

Term	Definition
'Legal metrological control'	means the control exercised by a public authority or regulator of the measurement tasks intended for the field of application of a measuring instrument, for reasons of public interest, public health, public safety, public order, protection of the environment, the levying of taxes and duties, the protection of consumers and fair trading
'Measurable heat'	means a net heat flow transported through identifiable pipelines or ducts using a heat transfer medium, such as, in particular, steam, hot air, water, oil, liquid metals and salts, for which a heat meter is or could be installed
'Measurement point'	means the emission source for which continuous emission measurement systems (CEMS) are used for emission measurement, or the cross-section of a pipeline system for which the CO_2 flow is determined using continuous measurement systems
'Measurement system'	means a complete set of measuring instruments and other equipment used to determine variables for the monitoring and calculation of emissions
'Minimum requirements'	means monitoring methods using the minimum efforts allowed for determining data in order to result in emission data acceptable for the purpose of the CBAM Regulation.
'Mixed fuel'	means a fuel which contains both biomass and fossil carbon
'Mixed material'	means a material which contains both biomass and fossil carbon
'Multifunctional process'	means a process that delivers multiple outputs or whose outputs feed into several production processes
'Net calorific value' (NCV)	means the specific amount of energy released as heat when a fuel or material undergoes complete combustion with oxygen under standard conditions, less the heat of vaporisation of any water formed
'Non-CBAM good'	means any good produced in the installation that is not included in Annex I to the CBAM Regulation
'Non-measurable heat'	means all heat other than measurable heat
'Operator'	means any person who operates or controls an installation in a third country, including a parent company that controls an installation in a third country
'Oxidation factor'	means the ratio of carbon oxidised to CO_2 as a consequence of combustion to the total carbon contained in the fuel, expressed as a fraction, considering carbon monoxide (CO) emitted to the atmosphere as the molar equivalent amount of carbon dioxide (CO_2)
'Power purchase agreement'	means a contract under which a person agrees to purchase electricity directly from an electricity producer

Term	Definition
'Precursor'	means any input material into a production process included in the list of goods set out in Annex I to the CBAM Regulation;
'Preliminary emission factor'	means the assumed total emission factor of a fuel or material based on the carbon content of its biomass fraction and its fossil fraction before multiplying it by the fossil fraction to produce the emission factor
'Production process'	means the chemical and physical processes carried out to produce goods in an installation
'Production route' [9]	means a specific technology used in a production process to produce goods
'Process emissions'	means greenhouse gas emissions other than combustion emissions occurring as a result of intentional and unintentional reactions between substances or their transformation, for a primary purpose other than the generation of heat, including from the following processes: (a) the chemical, electrolytic or pyrometallurgical reduction of metal compounds in ores, concentrates and secondary materials; (b) the removal of impurities from metals and metal compounds; (c) the decomposition of carbonates, including those used for flue gas cleaning; (d) chemical syntheses of products and intermediate products where the carbon bearing material participates in the reaction; (e) the use of carbon containing additives or raw materials; (f) the chemical or electrolytic reduction of metalloid oxides or non-metal oxides such as silicon oxides and phosphates.
'Proxy data'	means annual values which are empirically substantiated or derived from accepted sources and which an operator uses to substitute a data set[10] for the purpose of ensuring complete reporting
'Rebate'	means any amount that reduces the amount due or paid by a person liable for the payment of a carbon price, before its payment or after, in a monetary form or in any other form.
'Recommended improvements'	means monitoring methods which are proven means to ensure that data are more accurate or less prone to mistakes than by mere application of minimum requirements
'Related emissions'	means: – the direct emissions caused by the source stream or emission source concerned; – emissions attributed to a quantity of measurable heat; – the indirect emissions related to the quantity of electricity concerned; – embedded emissions of a material produced or of a precursor consumed.

[9] Note that different production routes can fall within the same production process.

[10] Refers to the activity data or the calculation factors.

Term	Definition
'Reporting period'	means the period corresponding to the calendar year during which the good was produced and used by the authorised CBAM declarant as reference for the determination of embedded emissions
'Residue'	means a substance that is not the end product that a production process directly seeks to produce; it is not a primary aim of the production process and the process has not been deliberately modified to produce it
'Simple goods'	means goods produced in a production process requiring exclusively input materials (precursors) and fuels having zero embedded emissions
'Source stream'	means either of the following: (a) a specific fuel type, raw material or product giving rise to emissions of relevant greenhouse gases at one or more emission sources as a result of its consumption or production; (b) a specific fuel type, raw material or product containing carbon and included in the calculation of greenhouse gas emissions using a mass-balance method
'Specific embedded emissions'	means the embedded emissions of one tonne of goods, expressed as tonnes of CO_2e emissions per tonne of goods
'Standard conditions'	means temperature of 273,15 K and pressure conditions of 101 325 Pa defining normal cubic metres (Nm^3)
'System boundary'	means the group of chemical or physical processes included in the calculation of embedded emissions of goods under the same aggregated goods category. The system boundaries shall cover direct emissions, indirect emissions for goods not listed in Annex II to the CBAM Regulation, and the embedded emissions of any precursor.
'Third country'	means a country or territory outside the customs territory of the [European] Union
'Tonne of $CO_2(e)$'	means one metric tonne of carbon dioxide ('CO_2'), or an amount of any other greenhouse gas listed in Annex I [to the CBAM Regulation] with an equivalent global warming potential ['CO_2e']
'Transmission system operator'	means an operator as defined in Article 2, point (35) of Directive (EU) 2019/944 of the European Parliament and of the Council ([11]).

[11] Directive (EU) 2019/944 of the European Parliament and of the Council of 5 June 2019 on common rules for the internal market for electricity and amending Directive 2012/27/EU (OJ L 158, 14.6.2019, p. 125).

Term	Definition
'Uncertainty'	means a parameter, associated with the result of the determination of a quantity, that characterises the dispersion of the values that could reasonably be attributed to the particular quantity, including the effects of systematic as well as of random factors, expressed in percent, and describes a confidence interval around the mean value comprising 95% of inferred values taking into account any asymmetry of the distribution of values
'Waste'	means any substance or object which the holder discards or intends or is required to discard, excluding substances that have been intentionally modified or contaminated in order to meet this definition
'Waste gas'	means a gas containing incompletely oxidised carbon in a gaseous state under standard conditions which is a result of any of the processes listed under 'process emissions'

Annex B List of abbreviations

Abbreviation	Full term
AD	Activity Data
AEM	Anode Effect Minutes
AEO	Anode Effect Overvoltage
AL	Activity Level
AOD	Argon Oxygen Decarburisation.
BAT	Best Available Techniques
BF	Biomass Fraction
BFG	Blast Furnace Gas
BM	Benchmark
BOF	Basic Oxygen Furnace
BOFG	Basic Oxygen Furnace Gas
BREFs	Best Available Techniques Reference documents
CA	Competent Authority
CBAM	Carbon Border Adjustment Mechanism, the
CCR	Clinker to Cement Ratio
CCS	Carbon Capture and Storage
CCU(S)	Carbon Capture and Utilisation (and Storage)
CEMS	Continuous Emission Measurement Systems
CF	Conversion Factor
CFP	Carbon footprint of products
CHP	Combined Heat and Power
CKD	Cement Kiln Dust
CN	Combined nomenclature
COG	Coke Oven Gas
DRI	Direct Reduced Iron
EAF	Electric Arc Furnace
EEA	European Economic Area (EU-27, Iceland, Norway, Liechtenstein)
EF	Emission Factor
EFTA	European Free Trade Area
EORI	Economic Operator Registration and Identification
ETS	Emissions Trading System
EU ETS	EU Emissions Trading System
EUA	EU Allowances (used in the EU ETS)
EUR	Euro (currency)

Abbreviation	Full term
FAR	Free Allocation Rules (Regulation 2019/331)[12]
FIFO	First-In-First-Out principle
GHG	Greenhouse Gas
GWP	Global Warming Potential
HBI	Hot Briquetted Iron
HS	Harmonised System (for international trade)
IEA	International Energy Agency
ISO	International Organization for Standardization
LULUCF	Land-use, land change and forestry (criteria)
MP	Monitoring Plan
MRR	Monitoring and Reporting Regulation (Regulation 2018/2066)[13]
MRV	Monitoring, Reporting and Verification
MS	Member State(s)
MWh	Megawatt-hour
NCV	Net Calorific Value
NPI	Nickel pig iron
OF	Oxidation Factor
PCI	Pulverised Coal Injection
PEMS	Predictive Emission Monitoring System
PFC	Perfluoro-carbon
PoS	Proofs of Sustainability
RED II	Renewable Energy Directive, re-cast
SEE	Specific embedded emissions
SEFA	Specific embedded free allocation
TARIC	Integrated Tariff of the European Union database
TJ	Terajoules
TSO	Transmission System Operator
UCC	Union Custom Code
UN/LOCODE	United Nations Code for Trade and Transport Location

[12] Free Allocation Rules (Commission Delegated Regulation (EU) 2019/331 of 19 December 2018 determining transitional Union-wide rules for harmonised free allocation of emission allowances pursuant to Article 10a of Directive 2003/87/EC of the European Parliament and of the Council)

[13] Monitoring and Reporting Regulation (Commission Implementing Regulation (EU) 2018/2066 of 19 December 2018 on the monitoring and reporting of greenhouse gas emissions pursuant to Directive 2003/87/EC of the European Parliament and of the Council and amending Commission Regulation (EU) No 601/2012